Assignment_2

```
mysql> CREATE DATABASE company_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE company_db;
Database changed
mysql> CREATE TABLE Department (
    ->     DeptID INT PRIMARY KEY,
    ->     DeptName VARCHAR(50) NOT NULL,
    ->     Location VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Employee (
    ->     EmpID INT PRIMARY KEY,
    ->     EmpName VARCHAR(100) NOT NULL,
    ->     DeptID INT,
    ->     Salary DECIMAL(10, 2),
    ->     JoiningDate DATE,
    ->     City VARCHAR(50),
    ->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> INSERT INTO Department (DeptID, DeptName, Location) VALUES
    -> (10, 'HR', 'Mumbai'),
    -> (20, 'IT', 'Pune'),
    -> (30, 'Sales', 'Mumbai');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Employee (EmpID, EmpName, DeptID, Salary, JoiningDate, City)
VALUES
    -> (1, 'Amit', 10, 45000.00, '2023-01-15', 'Mumbai'),
    -> (2, 'Priya', 20, 60000.00, '2022-03-20', 'Pune'),
    -> (3, 'Rahul', 30, 52000.00, '2021-07-10', 'Mumbai'),
    -> (4, 'Sneha', 20, 75000.00, '2024-02-01', 'Nagpur'),
    -> (5, 'Karan', 10, 48000.00, '2023-11-05', 'Nashik');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Employee ADD COLUMN Email VARCHAR(100);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Employee (EmpID, EmpName, DeptID, Salary, JoiningDate, City,
Email) VALUES
    -> (101, 'Aniket', 20, 58000.00, '2024-05-12', 'Pune', 'aniket@gmail.com'),
    -> (102, 'Pooja', 30, 62000.00, '2023-09-18', 'Mumbai', 'pooja@gmail.com'),
    -> (103, 'Suresh', 10, 49000.00, '2022-12-01', 'Thane', 'suresh@gmail.com'),
    -> (104, 'Neha', 20, 71000.00, '2024-01-10', 'Mumbai', 'neha@gmail.com'),
    -> (105, 'Rohan', 30, 53000.00, '2023-04-25', 'Pune', 'rohan@gmail.com');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> UPDATE Employee SET Salary = 65000.00 WHERE EmpID = 5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> CREATE USER IF NOT EXISTS 'readonly_user'@'localhost' IDENTIFIED BY
'password123';
Query OK, 0 rows affected, 1 warning (0.05 sec)

mysql> GRANT SELECT ON company_db.Employee TO 'readonly_user'@'localhost';
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER IF NOT EXISTS 'temp_user'@'localhost' IDENTIFIED BY
'password123';
Query OK, 0 rows affected, 1 warning (0.06 sec)

mysql> GRANT INSERT ON company_db.Employee TO 'temp_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE INSERT ON company_db.Employee FROM 'temp_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Employee (EmpID, EmpName, DeptID, Salary, JoiningDate, City)
    -> VALUES (106, 'Karan Deshmukh', 20, 68000.00, '2026-04-01', 'Mumbai');
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Employee SET Salary = 60000.00 WHERE EmpID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT sp_before_dept_update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Employee SET DeptID = 30 WHERE EmpID = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> ROLLBACK TO sp_before_dept_update;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Employee WHERE Salary > 50000;
+-------+---------------+--------+----------+-------------+--------+--------------
---+
| EmpID | EmpName       | DeptID | Salary   | JoiningDate | City   | Email
  |
+-------+---------------+--------+----------+-------------+--------+--------------
---+
|     2 | Priya         |     20 | 60000.00 | 2022-03-20  | Pune   | NULL
  |
|     3 | Rahul         |     30 | 52000.00 | 2021-07-10  | Mumbai | NULL
  |
|     4 | Sneha         |     20 | 75000.00 | 2024-02-01  | Nagpur | NULL
  |
|     5 | Karan         |     10 | 65000.00 | 2023-11-05  | Nashik | NULL
  |
|   101 | Aniket        |     20 | 60000.00 | 2024-05-12  | Pune   |
aniket@gmail.com |
|   102 | Pooja         |     30 | 62000.00 | 2023-09-18  | Mumbai |
pooja@gmail.com  |
|   104 | Neha          |     20 | 71000.00 | 2024-01-10  | Mumbai | neha@gmail.com
  |
```

```
|   105 | Rohan          |     30 | 53000.00 | 2023-04-25  | Pune   |
rohan@gmail.com  |
|   106 | Karan Deshmukh |     20 | 68000.00 | 2026-04-01  | Mumbai | NULL
  |
+-------+----------------+--------+----------+-------------+--------+--------------
---+
9 rows in set (0.00 sec)

mysql> UPDATE Employee SET Salary = Salary * 1.10 WHERE DeptID = 10;
Query OK, 3 rows affected (0.00 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> SELECT DeptID, COUNT(*) AS TotalEmployees FROM Employee GROUP BY DeptID;
+--------+----------------+
| DeptID | TotalEmployees |
+--------+----------------+
|     10 |              3 |
|     20 |              5 |
|     30 |              3 |
+--------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT d.DeptID, d.DeptName
    -> FROM Department d
    -> LEFT JOIN Employee e ON d.DeptID = e.DeptID
    -> WHERE e.EmpID IS NULL;
Empty set (0.00 sec)

mysql>
mysql> DROP VIEW IF EXISTS MumbaiSalespeople;
Query OK, 0 rows affected, 1 warning (0.02 sec)

mysql> CREATE VIEW MumbaiSalespeople AS
    -> SELECT * FROM Employee WHERE City = 'Mumbai' AND DeptID = 30;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT MAX(Salary) AS HighestSalary FROM Employee;
+---------------+
| HighestSalary |
+---------------+
|      75000.00 |
+---------------+
1 row in set (0.00 sec)

mysql> SELECT DISTINCT Salary FROM Employee ORDER BY Salary DESC LIMIT 1 OFFSET 1;
+----------+
| Salary   |
+----------+
| 71500.00 |
+----------+
1 row in set (0.00 sec)

mysql> SELECT * FROM Employee ORDER BY JoiningDate DESC;
+-------+----------------+--------+----------+-------------+--------+--------------
---+
| EmpID | EmpName        | DeptID | Salary   | JoiningDate | City   | Email
  |
+-------+----------------+--------+----------+-------------+--------+--------------
---+
|   106 | Karan Deshmukh |     20 | 68000.00 | 2026-04-01  | Mumbai | NULL
  |
|   101 | Aniket         |     20 | 60000.00 | 2024-05-12  | Pune   |
aniket@gmail.com |
|     4 | Sneha          |     20 | 75000.00 | 2024-02-01  | Nagpur | NULL
  |
|   104 | Neha           |     20 | 71000.00 | 2024-01-10  | Mumbai | neha@gmail.com
  |
```

```
|     5 | Karan          |     10 | 71500.00 | 2023-11-05  | Nashik | NULL
  |
|   102 | Pooja          |     30 | 62000.00 | 2023-09-18  | Mumbai |
pooja@gmail.com  |
|   105 | Rohan          |     30 | 53000.00 | 2023-04-25  | Pune   |
rohan@gmail.com  |
|     1 | Amit           |     10 | 49500.00 | 2023-01-15  | Mumbai | NULL
  |
|   103 | Suresh         |     10 | 53900.00 | 2022-12-01  | Thane  |
suresh@gmail.com |
|     2 | Priya          |     20 | 60000.00 | 2022-03-20  | Pune   | NULL
  |
|     3 | Rahul          |     30 | 52000.00 | 2021-07-10  | Mumbai | NULL
  |
+-------+----------------+--------+----------+-------------+--------+--------------
---+
11 rows in set (0.00 sec)

mysql> ALTER TABLE Department RENAME COLUMN DeptName TO DepartmentName;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```